EE4047 – High Voltage Techniques I

Continuation to:

The Fundamentals of High Voltage Engineering

Week 3.1 - Monday Lesson

Asst. Prof. Dr. M. Alparslan

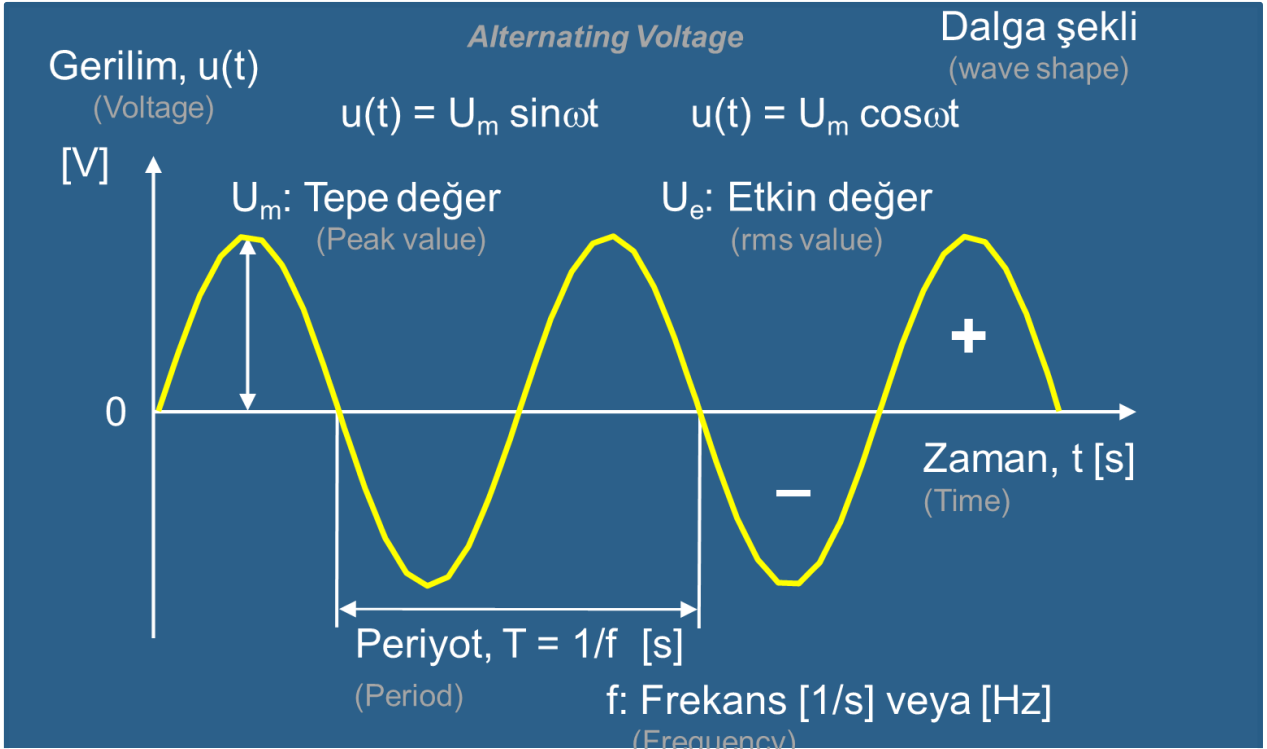
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OUTLINE

3 **Continuation to:** **The Fundamentals of High Voltage Engineering**

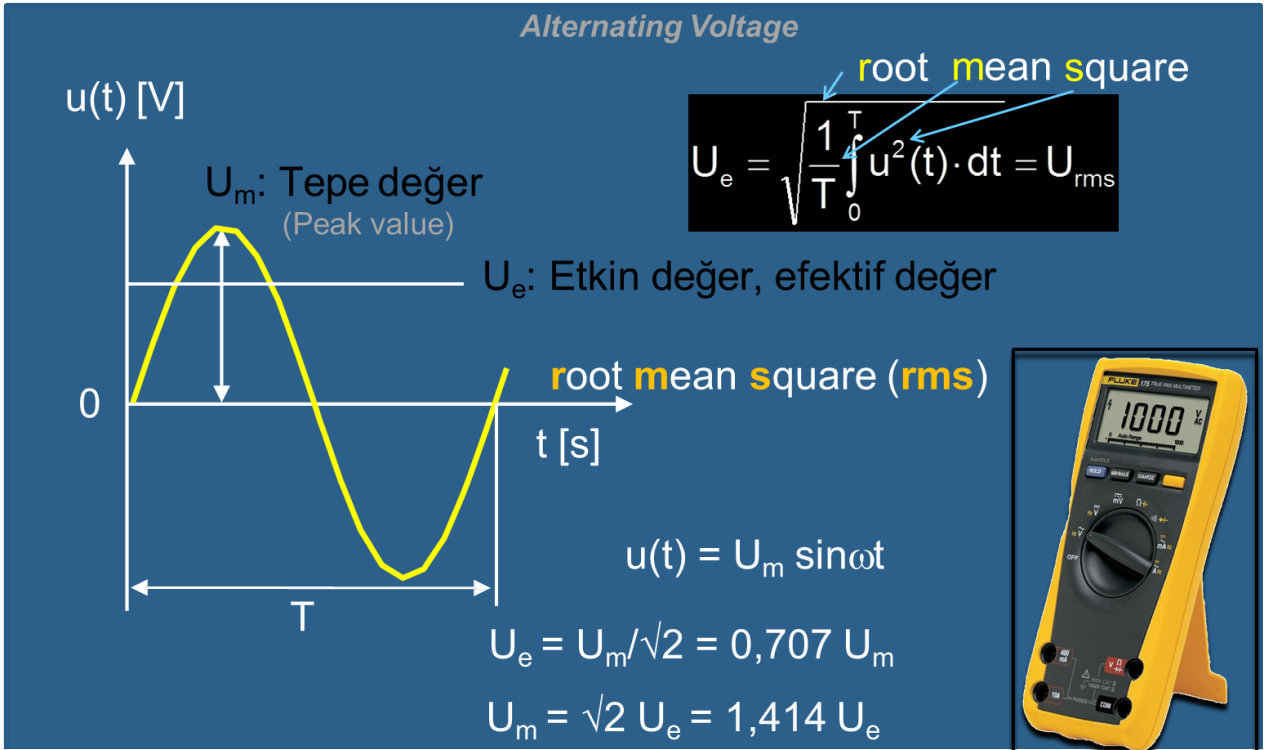
HV Concepts

Alternating Voltage:



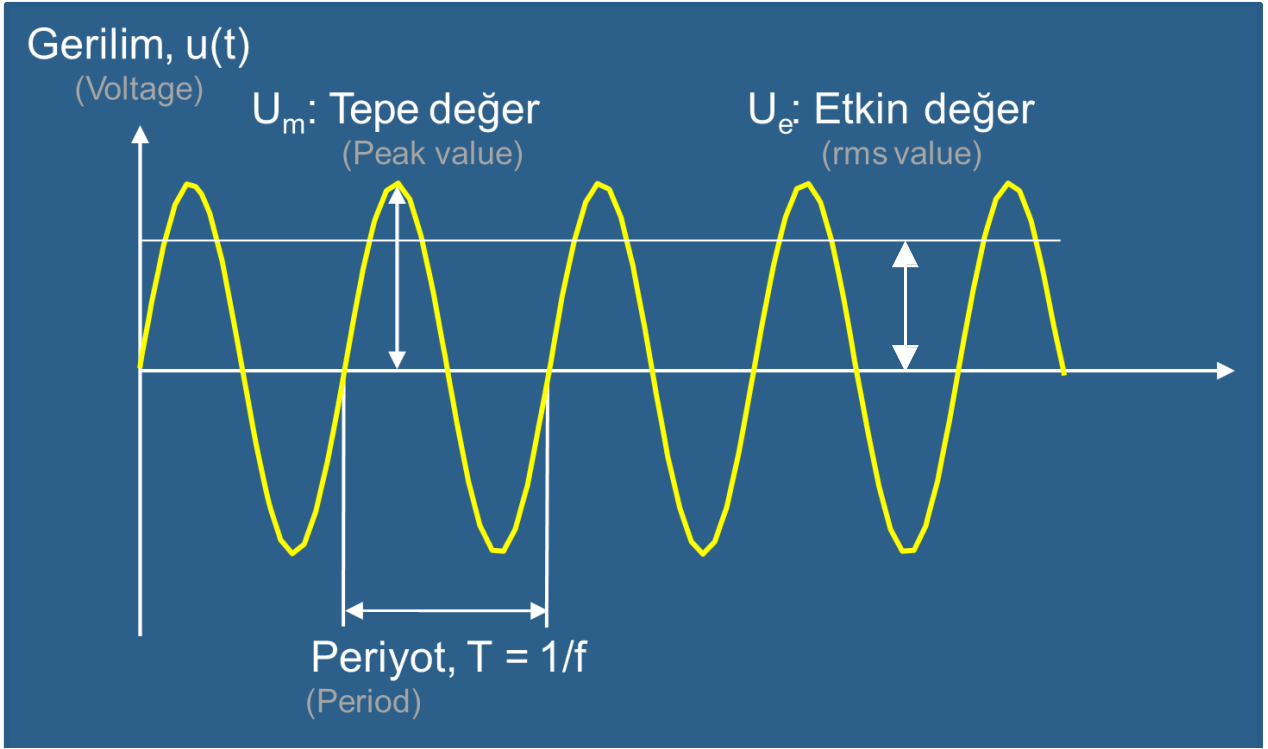
HV Concepts

Alternating Voltage:



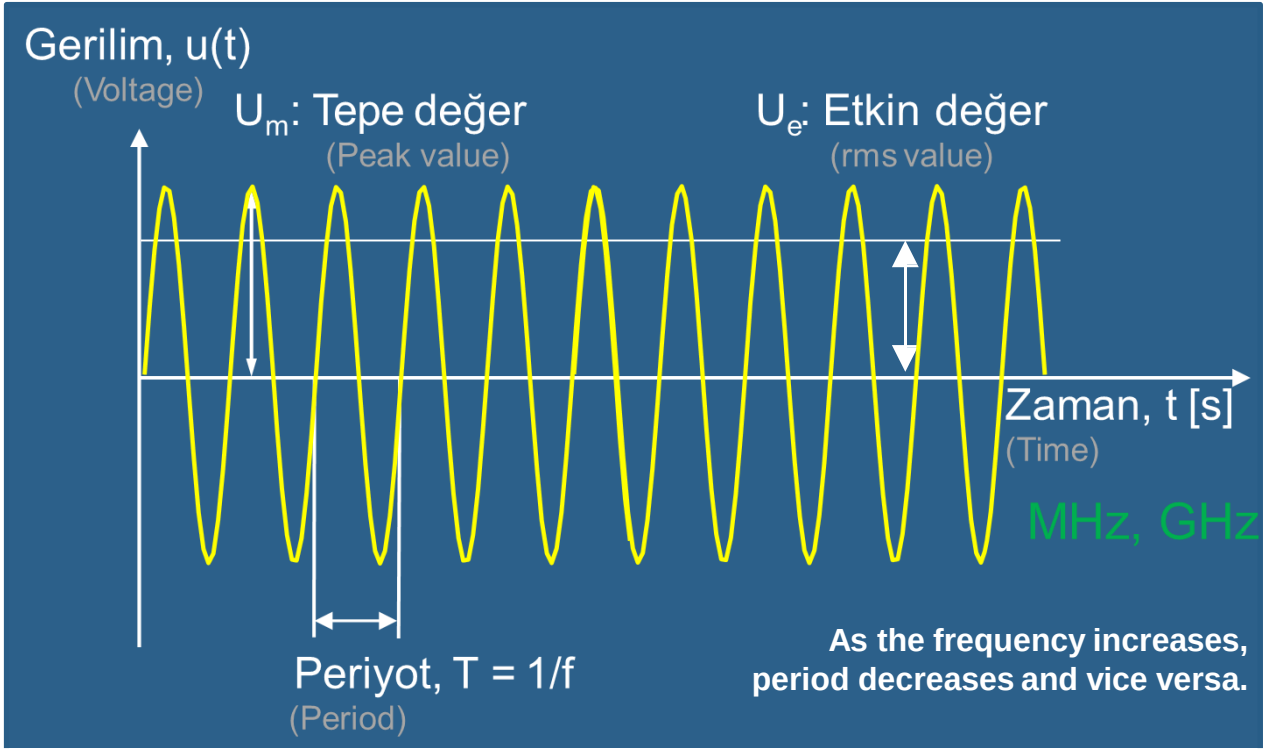
HV Concepts

Alternating Voltage:



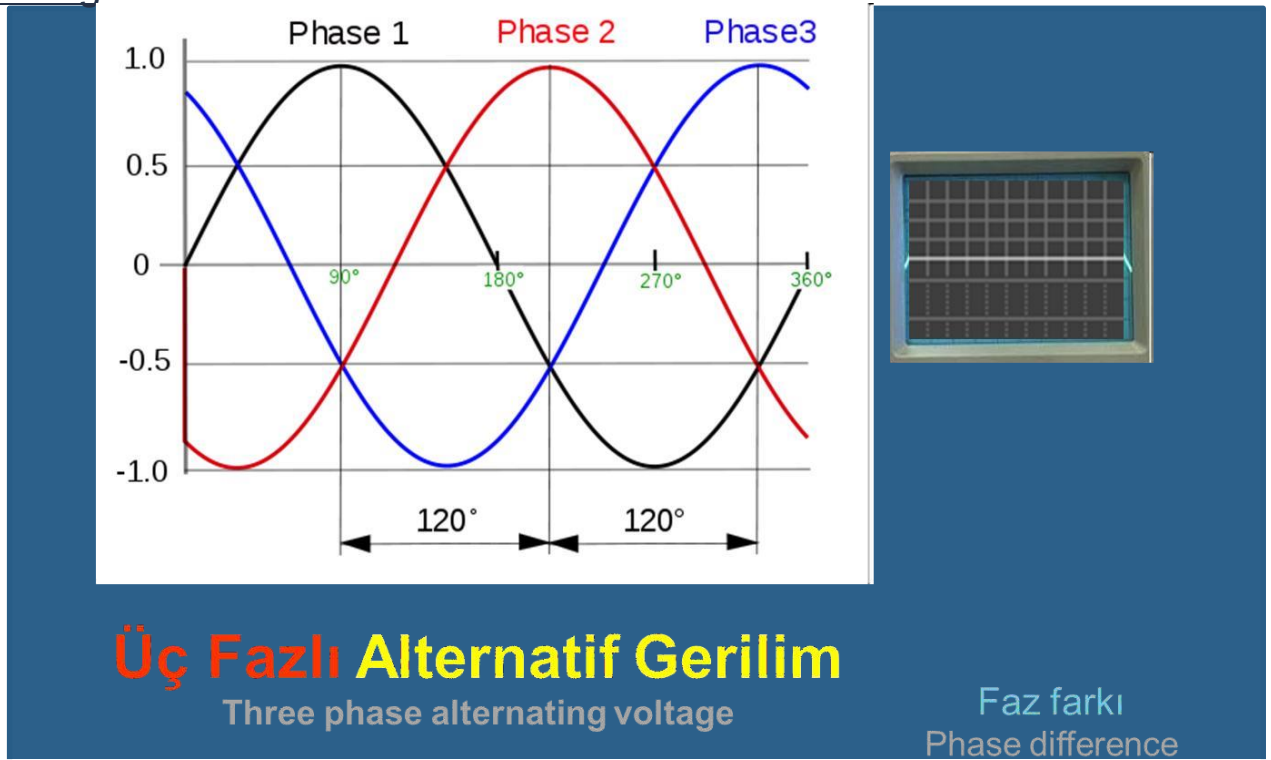
HV Concepts

Alternating Voltage:



HV Concepts

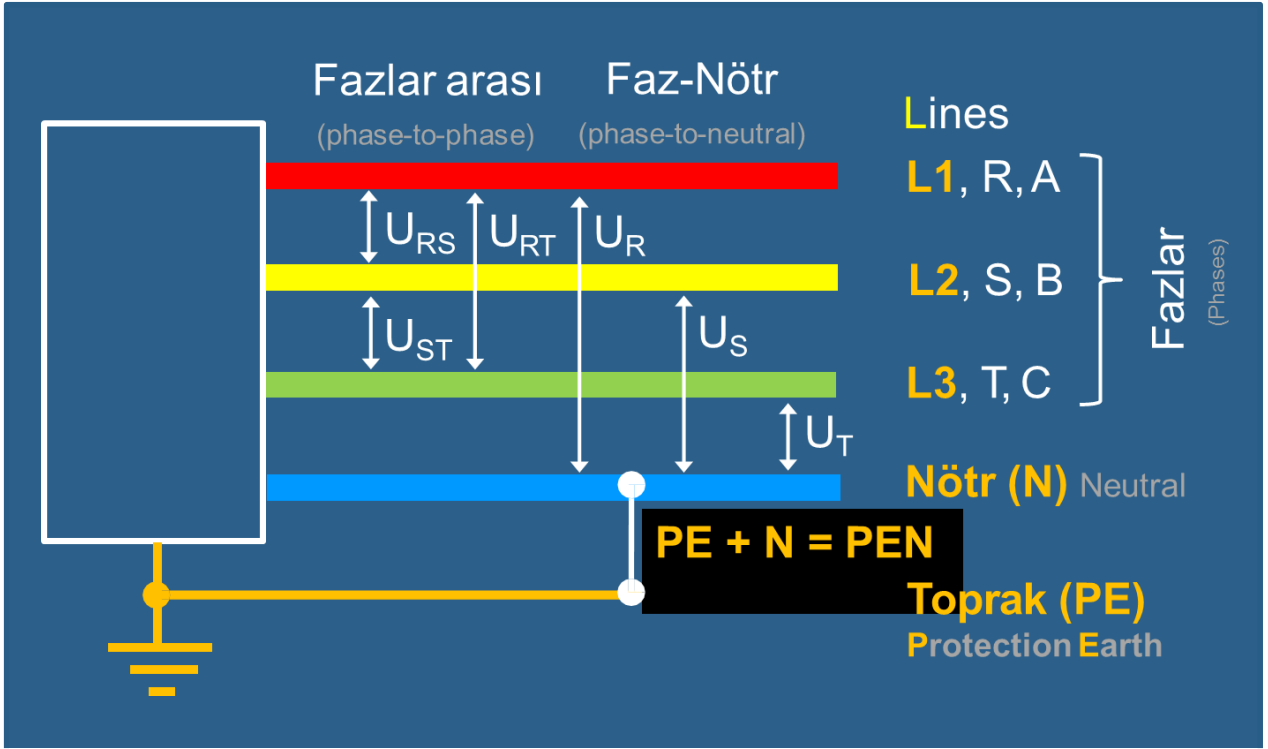
Alternating Voltage:



THE FUNDAMENTALS OF HIGH VOLTAGE ENGINEERING

HV Concepts

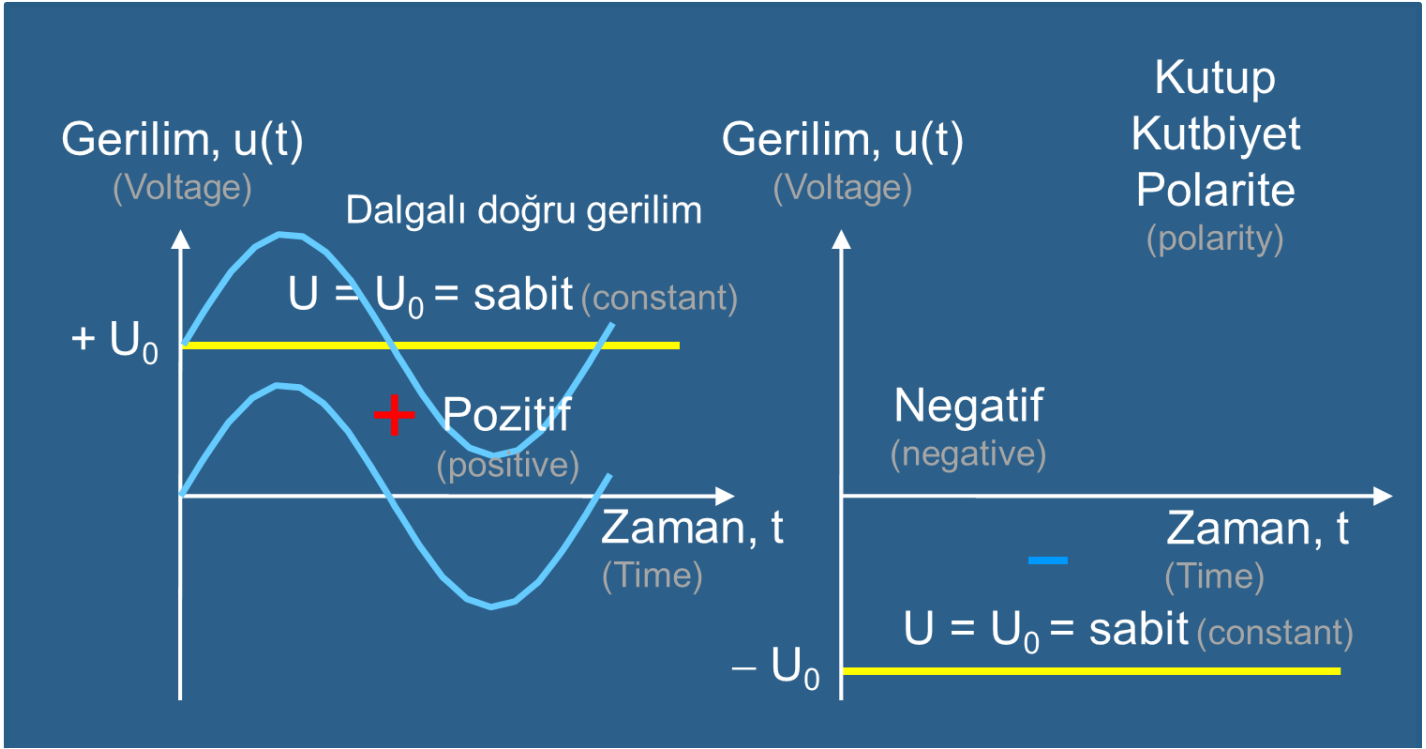
Alternating Voltage:



THE FUNDAMENTALS OF HIGH VOLTAGE ENGINEERING

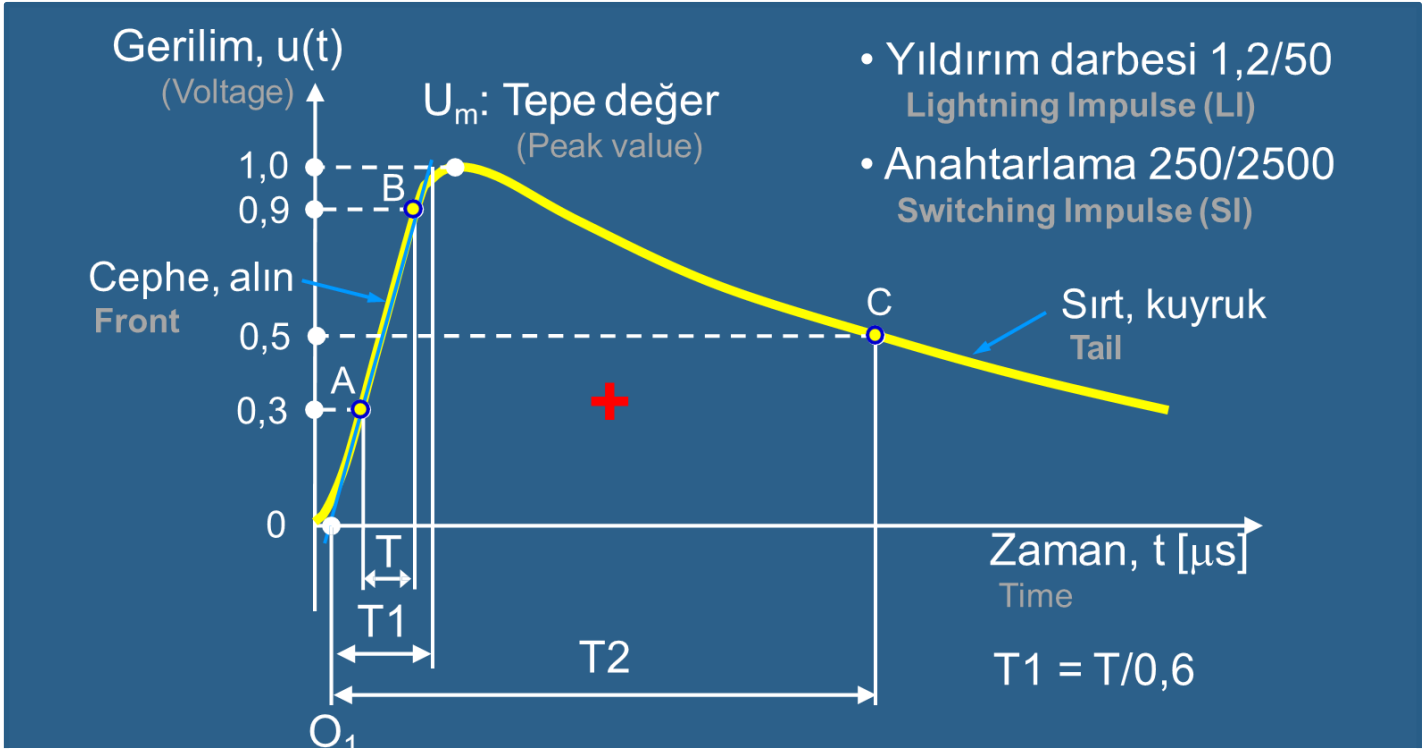
HV Concepts

Direct Voltage:



HV Concepts

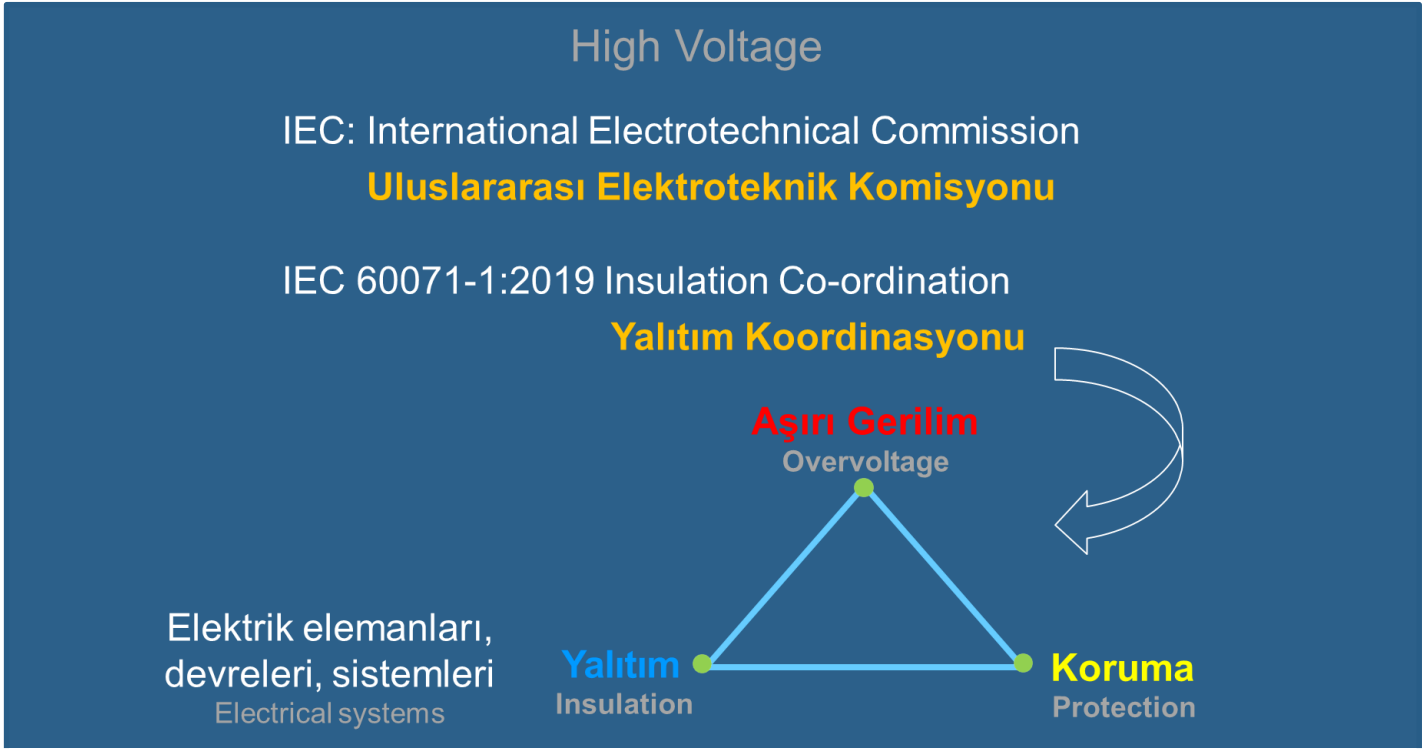
Impulse Voltage:



THE FUNDAMENTALS OF HIGH VOLTAGE ENGINEERING

HV Concepts

Related standard:



HV Concepts

Insulation and Conductivity:

- Işık (light)
- Ses (sound)
- Isı (heat)
- Su (water)
- Elektrik (electricity)
- Akım (current)
- Gerilim (voltage)
- Yalıtım (insulation)
- İletim (conduction)

- İletim (conduction)
- İletkenlik (conductivity)
- İletken (conductor)

- Yalıtım (insulation)
- Yalıtkanlık (electrical resistivity)
- Yalıtkan (insulator)



$$Current = \frac{Voltage}{Resistance}$$

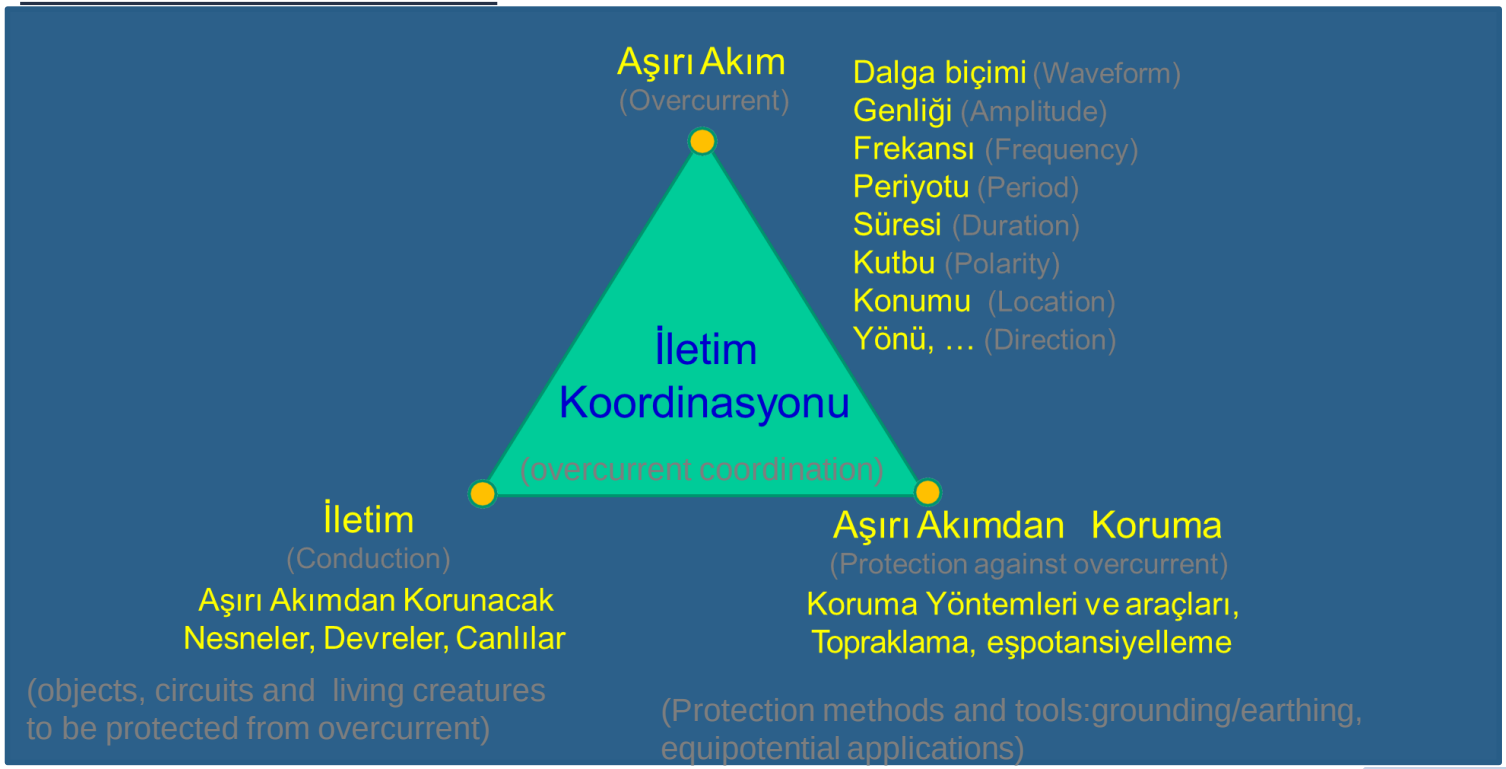
HV Concepts

Insulation Coordination:



HV Concepts

Conduction Coordination:



3 THE FUNDAMENTALS OF HIGH VOLTAGE ENGINEERING

The elements you will see here are for use at high voltage.



THE FUNDAMENTALS OF HIGH VOLTAGE ENGINEERING

A. Temel Elemanlar
Basic Elements

1. Generatörler (Generators)
2. Transformatörler (Trafos)
3. Kesiciler (Circuit-Breakers)
4. Ayırıcılar (Disconnectors)
5. Hatlar (Transmission Lines)
6. Direkler (Poles, Pylons)
7. İzolatörler (Isolators)
8. Kablolar (Cables)
9. Baralar (Buses)

B. Ölçme ve Koruma Elemanları
Measurement and Protection Elements

1. Akım Transformatörü (Current Transformer)
2. Gerilim Transformatörü (VoltageTransformer)
3. Gerilim Bölücüler (Dividers)
4. Parafudrlar (Lightning Arresters)
5. Ark Boynuzları (Arc Horns)
6. Koruma Halkaları (Guard ring)
7. Koruma Hatları (Grounding lines)
8. Röleler (Relays)

THE FUNDAMENTALS OF HIGH VOLTAGE ENGINEERING

C. Kontrol ve Kumanda Elemanları
Control and Operating Elements

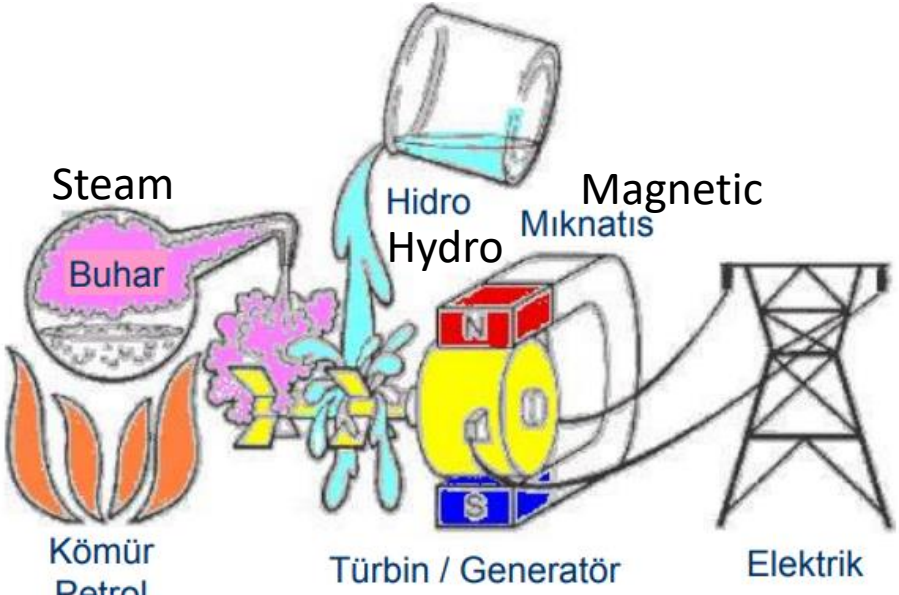
1. Röleler (Relays)
2. Kontrol ve Kumanda Devreleri (Control Circuits)

D. Gerilim Ayar Elemanları
Voltage Adjustment Elements

1. Seri ve Paralel Bağlı Reaktörler (Series and Parallel Connected Reactors)
2. Seri ve Paralel Bağlı Kondansatörler (Series and Parallel Connected Capacitors)
3. Regülatörler (Regulators)

THE FUNDAMENTALS OF HIGH VOLTAGE ENGINEERING

Electricity generation (conventional)



Coal
Petroleum
Natural
Gas

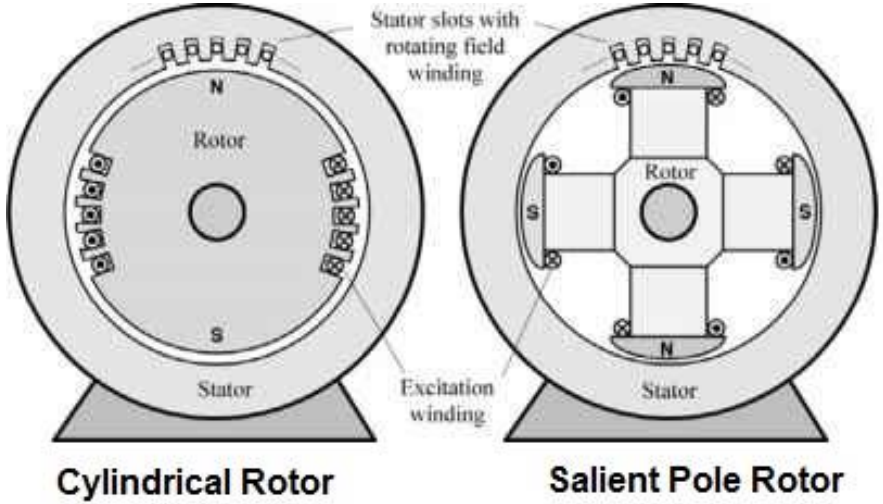
Turbine /
Generator

Electricity

THE FUNDAMENTALS OF HIGH VOLTAGE ENGINEERING

Synchronous Generators

1. Turbo Generators
2. Salient pole Generators



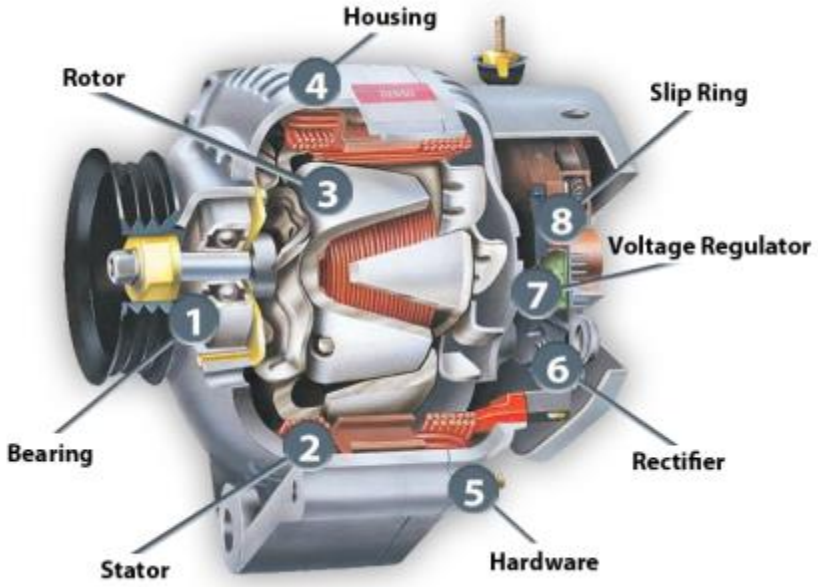
THE FUNDAMENTALS OF HIGH VOLTAGE ENGINEERING

Synchronous Generators



THE FUNDAMENTALS OF HIGH VOLTAGE ENGINEERING

Synchronous Generators



THE FUNDAMENTALS OF HIGH VOLTAGE ENGINEERING

Synchronous Generators



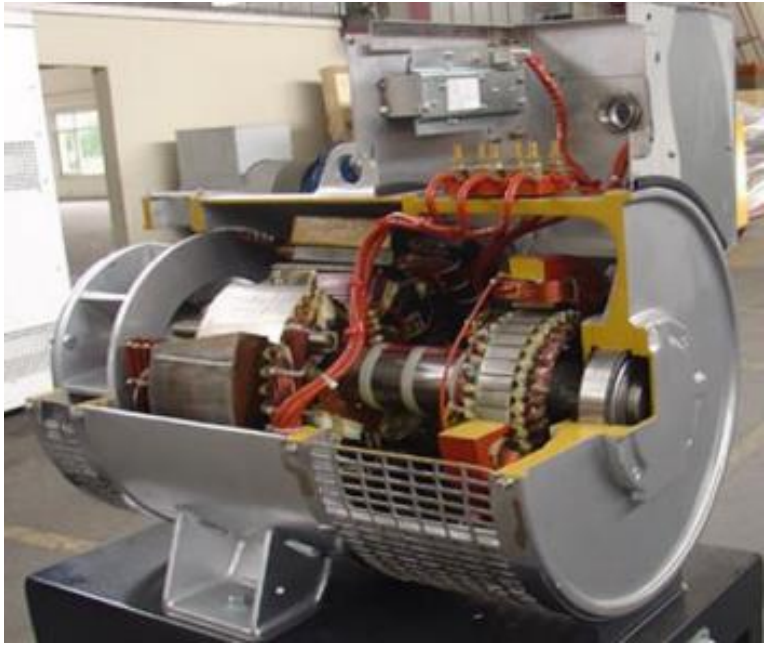
THE FUNDAMENTALS OF HIGH VOLTAGE ENGINEERING

Synchronous Generators



THE FUNDAMENTALS OF HIGH VOLTAGE ENGINEERING

Synchronous Generators



THE FUNDAMENTALS OF HIGH VOLTAGE ENGINEERING

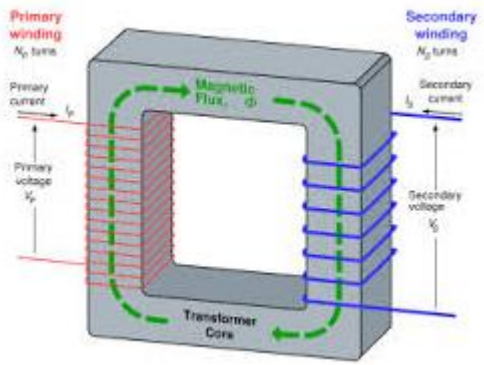
Synchronous Generators



THE FUNDAMENTALS OF HIGH VOLTAGE ENGINEERING

Power Transformers

Static electrical machines. Transform AC voltage level to a different level based on the ratio of primary and secondary windings. They have high efficiency. Usually produced for single-phase and three-phase use purposes.





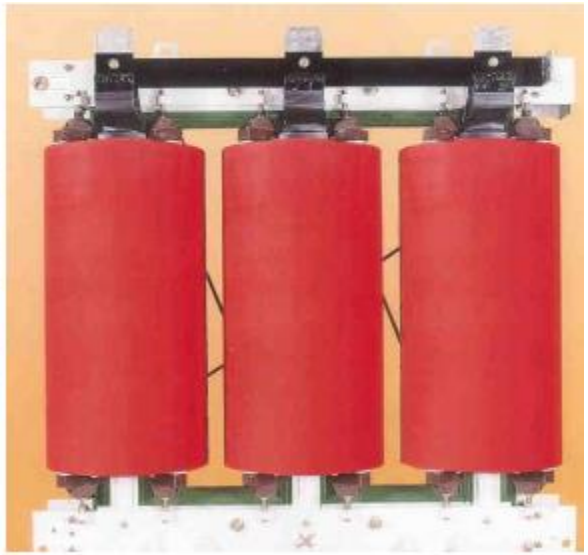


THE FUNDAMENTALS OF HIGH VOLTAGE ENGINEERING

Power Transformers

Dry type power trasnformer.





THE FUNDAMENTALS OF HIGH VOLTAGE ENGINEERING

Power Transformers

Oil type power trasnformer.



THE FUNDAMENTALS OF HIGH VOLTAGE ENGINEERING

Power Transformers





THE FUNDAMENTALS OF HIGH VOLTAGE ENGINEERING

Power Transformers

Test / experiment transformers (in the figure, 750 kV, 2100 kVA)



THE FUNDAMENTALS OF HIGH VOLTAGE ENGINEERING

Power Transformers

Test / experiment transformers (in the figure, 1500 kV, 1.2 A, 4 stage cascaded)



STUDY QUESTIONS: Fundamentals of High Voltage Engineering

1. What are the main parts of interconnected power systems?
2. What is peak demand, how it changes annually in Turkey?
3. What is installed power, how is it compared to peak demand?
4. Where are the hydropower, thermal, wind and geothermal power plants mostly located geographically in Turkey?
5. What are the four sources with the highest share in electricity generation in Turkey?
6. Compare the overall length of transmission lines with distribution lines in Turkey.
7. How and why high voltage started to be commonly preferred for which reasons in the history of electricity?
8. How electric power that can be transmitted on an overhead AC line changes with voltage?
9. Why we do not prefer much higher voltages than the ones we are using today?
10. What are the origins of overvoltages?

STUDY QUESTIONS: Fundamentals of High Voltage Engineering

11. Where and why HVDC is preferred?

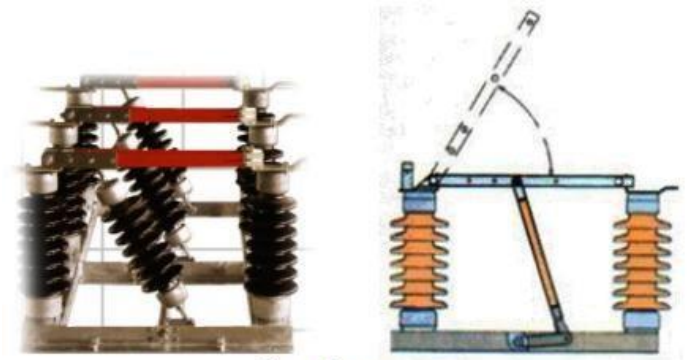
12. Which voltages are high voltage? What are the subcategories of high voltage?

13. What are the types of HV components? Give three examples for each type.

High Voltage Elements

Disconnectors (Isolators)

The switchgear capable of openning or closing the circuit **when there is no current flow** in MV and HV systems.



Bina dışı ayırıcı
(Outdoor Disconnector)

When they are opened, they leave **a visible gap** between the contacts. In applications they are also known as **sectionalizer**. Recently they are more popularly called as **disconnector**.

They disconnect different parts of the electrical circuit from each other allowing **safe maintenance and control** activities. Moreover, they are used in circuits with several main busses for energizing and deenergizing procedures and coupling operations.

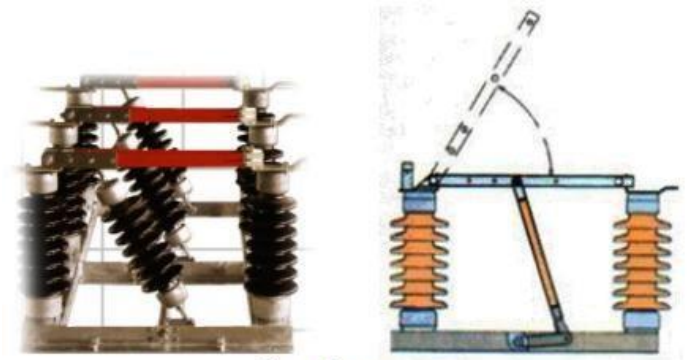
Disconnectors can not open or close when there is current in the circuit. If tried to do so, the disconnector and the person controlling it can get damage.

High Voltage Elements

Disconnectors

An example openning precedure:

- First, circuit breaker is opened.
- Then the disconnectors in both sides of the circuit breaker are opened.



Bina dışı ayırıcı
(Outdoor Disconnector)

When closing the circuit, these are done in reverse order:

- First, the disconnectors are closed.
- Then the circuit breaker is closed and the circuit is energized.

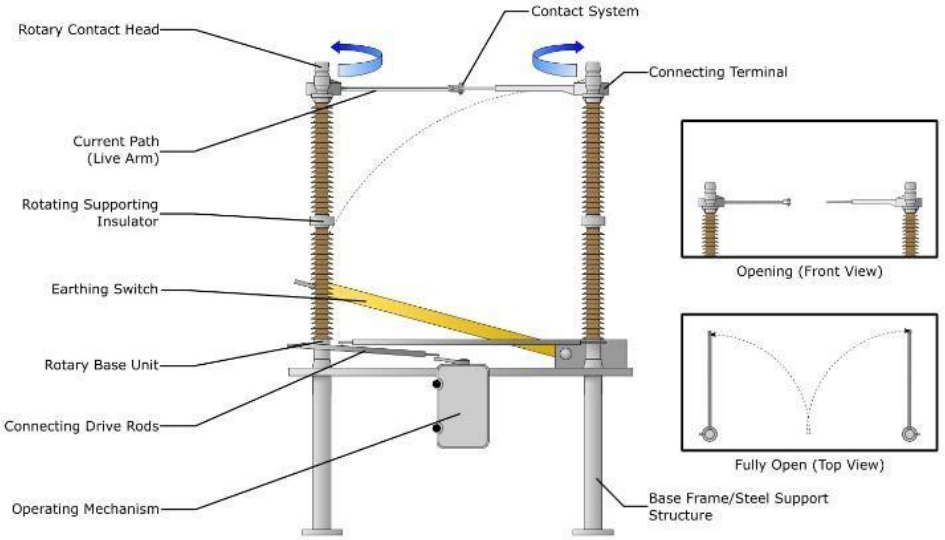
Note: If there are no circuit breakers in the system, first the loads are switched off, then disconnector can be opened.

THE FUNDAMENTALS OF HIGH VOLTAGE ENGINEERING

High Voltage Elements

Disconnectors (Isolators) Parts

of a typical disconnector:



- Current/live part: The part of the Disconnector that carries electrical load/current.
- Contact system: The point where the disconnector makes or breaks the electrical circuit.
- Supporting and rotating insulators: insulators reduce leakage/creepagecurrent and reduce the likelihood of flashover.
- Operating drive and connecting rods: Used to actuate (make/break) the disconnectors (usually an electric motor is used)
- Base frame: allows for the disconnector to be mounted or installed easily.

High Voltage Elements

Disconnectors (Isolators)

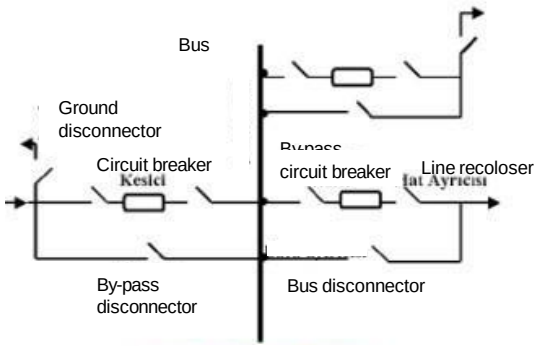
Disconnectors are classified according to four aspects:

- Their purpose of use
- The place that they are mounted
- Their structural aspects
- Their control types

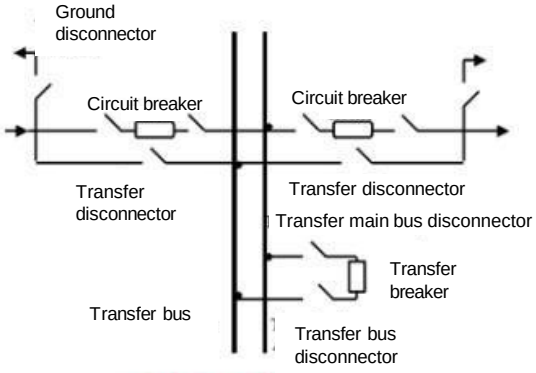
3 THE FUNDAMENTALS OF HIGH VOLTAGE ENGINEERING

- Bus disconnector
- Earth/ground disconnector
- By-pass disconnector
- Transfer disconnector
- Bus sectionalizing disconnector

-



SINGLE BUS SYSTEM



TWO-BUS SYSTEM

High Voltage Elements

Disconnectors (Isolators)

Classification due to t**he place that they are mounted**:

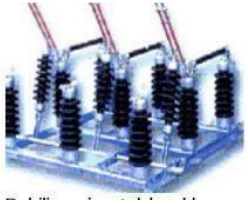
- Internal disconnectors (in closed spaces and substations)
- External disconnectors (in open substations and over poles)

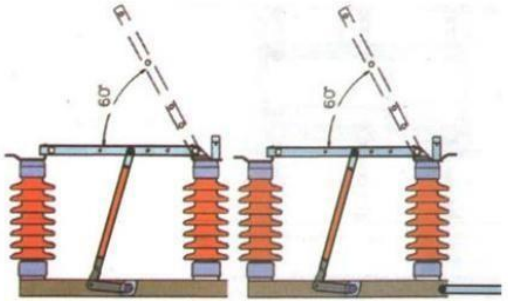
High Voltage Elements

Disconnectors (Isolators)

Classification due to **their structural aspects**:

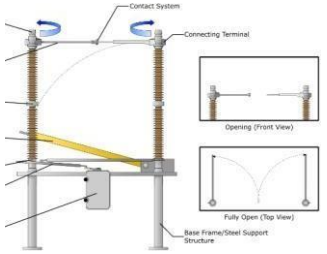
- Knife switch disconnector





  - internal, external, ground, fused versions
- Rotating disconnector





- Load break switch (can break rated current in the air or in a vacuum chamber, can not break short circuit current)



High Voltage Elements

Disconnectors (Isolators)

Classification due to **their control types**:

- Hand controlled disconnectors
- Mechanically controlled disconnectors
- Electric motor controlled disconnectors
- Pneumatically controlled disconnectors

High Voltage Elements

Circuit Breakers

Some fundamental information about circuit breaking:

-Load current is usually below rated (nominal current) of the system. It is the effective (RMS) value of the current that allows related equipment to be operated as long as desired.

- Overcurrent is violation of the nominal current value.

- Short-circuit current is caused due to a fault in the grid. Its value depends on generators, fault type and network impedance.

High Voltage Elements

Circuit Breakers

Additionally, circuit breakers are subject to several impacts when making, breaking or operating:

- Dielectric impacts(due to voltage)
- Thermal impacts (due to nominal and fault currents)
- Electrodynamic impacts (due to fault currents)
- Mechanical impacts.

The biggest impacts take place during circuit making and breaking actions. These cause arc phenomena which is usually hard to model and predict accurately.

Experience and experimental activities play big role in equipment design. They are usually electromechanical equipment. Static equipment is not technically and economically suitable in most of the medium and high voltage applications.

High Voltage Elements

Circuit Breakers

The ability to make or breake circuit at certain parts of electrical grid is required for grid operation and protection.

Circuit breakers are capable of openning or closing the circuit in no load, load and short circuit conditions.

Circuit breakers can carry the load, break the load and resist for the re-occurence of current.

Circuit breakers are,

- great conductors when they are closed,
- great isolators when they are opened,
- can handle openning, closing actions very fast.

END OF THE LESSON

Any questions?

Check out the Power Engineering playlist with 32 videos on Youtube:

https://www.youtube.com/watch?v=JwRTpWZReJk&list=PLLZI7WaEmZi3F134eDujF5ZNGxA1E0dkm